

Recall:

If $A = [v_1 \ \dots \ v_n]$ is an $m \times n$ matrix then:

- 1) $\text{Col}(A) = \text{Span}(v_1, \dots, v_n)$
- 2) $\text{Nul}(A) = \{v \in \mathbb{R}^n \mid Av = 0\}$

e.g. $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$

$$\text{Col}(A) = \text{Span} \left(\begin{bmatrix} 1 \\ 4 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \end{bmatrix}, \begin{bmatrix} 3 \\ 6 \end{bmatrix} \right) \subseteq \mathbb{R}^2$$

$$\text{Nul}(A) = \left\{ \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} \mid \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \cdot \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\} \subseteq \mathbb{R}^3$$

In general: $\text{Col}(A)$ is a subspace of $\mathbb{R}^m$
 $\text{Nul}(A)$ is a subspace of $\mathbb{R}^n$

Construction of a basis of $\text{Col}(A)$

Example

$$A = \begin{bmatrix} 1 & 2 & 5 \\ 2 & 3 & 8 \end{bmatrix}$$

$$\text{Col}(A) = \text{Span}\left(\underbrace{\begin{bmatrix} 1 \\ 2 \end{bmatrix}}_{v_1}, \underbrace{\begin{bmatrix} 2 \\ 3 \end{bmatrix}}_{v_2}, \underbrace{\begin{bmatrix} 5 \\ 8 \end{bmatrix}}_{v_3}\right) = \text{Span}\left(\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \end{bmatrix}\right)$$

$\uparrow$ $v_3 = v_1 + 2v_2$

Lemma

Let V be a vector space, and let $v_1, \dots, v_p \in V$. If a vector v_i is a linear combination of the other vectors then

$$\text{Span}(v_1, \dots, v_p) = \text{Span}(v_1, \dots, v_{i-1}, v_{i+1}, \dots, v_p)$$

Upshot. One can construct a basis of a vector space V as follows:

- Start with a set of vectors $\{v_1, \dots, v_p\}$ such that $\text{Span}(v_1, \dots, v_p) = V$.
- Keep removing vectors without changing the span, until you get a linearly independent set.

Example. Find a basis of $\text{Col}(A)$ where A is the following matrix:

$$A = \begin{matrix} & \begin{matrix} v_1 & v_2 & v_3 & v_4 & v_5 & v_6 \end{matrix} \\ \begin{matrix} \textcircled{1} \\ 0 \\ 0 \\ 0 \\ 0 \end{matrix} & \begin{bmatrix} 0 & 2 & 0 & 1 & 0 \\ \textcircled{1} & 3 & 0 & -1 & 0 \\ 0 & 0 & \textcircled{1} & 3 & 0 \\ 0 & 0 & 0 & 0 & \textcircled{1} \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

matrix in the reduced row echelon form

Solution:

$$\text{Col}(A) = \text{Span}(v_1, v_2, \cancel{v_3}, v_4, \cancel{v_5}, v_6) \subseteq \mathbb{R}^5$$

Note: $v_3 = 2v_1 + 3v_2$

$$v_5 = 1 \cdot v_1 + (-1)v_2 + 3v_4$$

This gives: $\text{Col}(A) = \text{Span}(v_1, v_2, v_4, v_6)$

The set $\{v_1, v_2, v_4, v_6\}$ is linearly independent, so it is a basis of $\text{Col}(A)$.

In general: If A is a matrix in the reduced row echelon form, then the set of all columns of A that contain leading ones is a basis of $\text{Col}(A)$.

Example. Find a basis of $\text{Col}(A)$ where A is the following matrix:

$$A = \begin{matrix} & \begin{matrix} v_1 & v_2 & v_3 & v_4 & v_5 \end{matrix} \\ \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix} \end{matrix}$$

Solution: $\text{Col}(A) = \text{Span}(v_1, v_2, v_3, v_4, v_5)$

$$\begin{matrix} \begin{matrix} v_1 & v_2 & v_3 & v_4 & v_5 \\ \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix} \end{matrix} & \xrightarrow{\text{row reduction}} & \begin{matrix} w_1 & w_2 & w_3 & w_4 & w_5 \\ \begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix} \end{matrix}$$

$$\begin{cases} v_2 = -2v_1 \\ v_4 = (-1)v_1 + 2v_3 \\ v_5 = 3v_1 - 2v_3 \end{cases} \quad \leftarrow \quad \begin{cases} w_2 = -2w_1 \\ w_4 = (-1)w_1 + 2w_3 \\ w_5 = 3w_1 - 2w_3 \end{cases}$$

This gives: $\text{Col}(A) = \text{Span}(v_1, v_3)$

Check: the set $\{v_1, v_3\}$ is linearly independent, so it is a basis of $\text{Col}(A)$.

In general: If A is a matrix then the set of pivot columns of A is a basis of $\text{Col}(A)$.

Construction of a basis of $\text{Nul}(A)$

Example. Find a basis of $\text{Nul}(A)$ where A is the following matrix:

$$A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

Solution:

Recall: We know how to find a spanning set of $\text{Nul}(A)$:

$$\left[\begin{array}{ccccc|c} -3 & 6 & -1 & 1 & -7 & 0 \\ 1 & -2 & 2 & 3 & -1 & 0 \\ 2 & -4 & 5 & 8 & -4 & 0 \end{array} \right] \xrightarrow{\text{row reduction}} \left[\begin{array}{ccccc|c} x_1 & x_2 & x_3 & x_4 & x_5 & \\ 1 & -2 & 0 & -1 & 3 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

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We get: $v \in \text{Nul}(A)$ if

$$v = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2x_2 + x_4 - 3x_5 \\ x_2 \\ -2x_4 + 2x_5 \\ x_4 \\ x_5 \end{bmatrix} = x_2 \cdot \underbrace{\begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}}_{w_1} + x_4 \cdot \underbrace{\begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix}}_{w_2} + x_5 \cdot \underbrace{\begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix}}_{w_3}$$

This gives: $\text{Nul}(A) = \text{Span}(w_1, w_2, w_3)$.

Note: The set obtained in this way is always linearly independent, so it is a basis of $\text{Nul}(A)$.

Upshot. If A is matrix then:

$\dim \text{Col}(A)$ = the number of pivot columns of A

$\dim \text{Nul}(A)$ = the number of non-pivot columns of A

Definition

If A is a matrix then:

- the dimension of $\text{Col}(A)$ is called the *rank* of A and it is denoted $\text{rank}(A)$
- the dimension of $\text{Nul}(A)$ is called the *nullity* of A .

Eg. If A is the matrix from the last example then:

$\dim \text{Col}(A) = 2$, so: $\text{rank } A = 2$

$\dim \text{Nul}(A) = 3$, so: $\text{nullity of } A = 3$

The Rank Theorem

If A is an $m \times n$ matrix then

$$\text{rank}(A) + \dim \text{Nul}(A) = n$$

Proof: $(\text{rank } A) + (\dim \text{Nul}(A))$

= (the number of pivot columns of A)

+ (the number of non-pivot columns of A)

= (the number of all columns of A)

= n

Low rank matrices in data analysis

	<i>exam 1</i>	<i>exam 2</i>	<i>exam 3</i>	<i>exam 4</i>
student 1	2	1	3	10
student 2	4	2	6	20
student 3	10	5	15	50
student 4	2	1	3	10
student 5	0	0	0	0
student 6	6	3	9	40